

Total No. of Questions : 10]

SEAT No. :

P3094

[Total No. of Pages : 5

[5461] - 131

B.E. (Mechanical Engineering)

REFRIGERATION AND AIR-CONDITIONING (Theory)

(2012 Pattern) (End Semester) (Semester - I) (402041)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer five questions from the following.*
- 2) *Draw neat labelled diagrams; wherever necessary.*
- 3) *Use of electronic calculator is permitted.*
- 4) *Use of Steam Table, Psychrometric chart, Refrigerant Property Table is Permitted.*
- 5) *Use of programmable calculator is not allowed.*
- 6) *Assume suitable/standard data, if necessary.*

Q1) a) Why Coefficient of Performance (COP) of a refrigerator is more during winter season than summer? **[4]**

- b) The Refrigeration cycle operates on Reverse Carnot Cycle. The capacity of a refrigerator is 200 TR, when working between 267 K and 298K. Determine the mass of ice produced per day from water at 25°C. Also; determine the power required to drive the unit?

Assume Latent heat of ice as 335 KJ/kg. ($L_{ice} = 335 \text{ kJ/kg}$). **[6]**

OR

Q2) a) Ethylene glycols and Propylene glycols are used as Secondary Refrigerants. Justify your answer? **[4]**

- b) A Bell-Coleman refrigeration cycle operates between pressures of 4 bar and 16 bar. The air temperature after heat rejection to surrounding is 37°C and air temperature at the exit of refrigerator is 7°C. The isentropic efficiencies of turbine and compressor are 0.85 and 0.80 respectively. Determine Compressor and Turbine work per ton of Refrigeration (TR) and COP of the system?

Assume $\gamma_{air} = 1.4$ and $C_{p-air} = 1.005 \text{ kJ/kg K}$. **[6]**

P.T.O.

- Q3) a)** What are the desirable properties of an ideal refrigerant? Also; enumerate any two thermodynamic properties required for the refrigerant? [4]
- b)** A multiple evaporator R22 refrigeration system has three evaporators, all working at the same temperature of -16°C between the temperature limits of -16°C and 30°C with single compressor. The refrigeration loads of respective evaporators are 20 TR, 10 TR and 30 TR respectively. The refrigerant vapour leaving the evaporators is dry and there is no sub-cooling of the refrigerant in the condenser. Determine: [6]
- Refrigeration effect per kg of the refrigerant.
 - Mass of the refrigerant circulated in kg/min in each evaporator.
 - Compressor power required.
 - COP of the system

Properties of R22

Saturation Temperature ($^{\circ}\text{C}$)	Specific heat of Vapour (KJ/kg K)	Specific Enthalpy KJ/kg		Specific Entropy (KJ/kg K)	
		h_f	h_g	s_f	s_g
-16°C	0.6782	26.72	245.08	0.01080	0.9576
30°C	0.9081	84.38	261.60	0.3115	0.8964.

OR

- Q4) a)** List out the two alternative refrigerants to CFC-12 in domestic refrigerators and justify your answer? [4]
- b)** In an Aqua-Ammonia vapour absorption refrigeration system; the heat is supplied to NH_3 generator by condensing steam at 2 bar and 98% dry. the temperature in the refrigerator is to be maintained at -5°C , when the surrounding temperature is at 30°C . Find the maximum possible COP of the system, if the saturation temperature of steam corresponding to 2 bar is 120°C and latent heat of steam is 2201 KJ/kg. [6]
- If the refrigeration load load is 20 TR and actual COP is 75% of the maximum COP, find the mass of steam required in kg/hr. Assume, only latent heat of steam is utilised for heating purpose?
- Q5) a)** Define the following terms: [4]
- Dalton's law of Partial Pressure.
 - Degree of Saturation.
 - Dew Point Temperature.
 - Saturated air.

- b) The pressure and temperature of the air in a room is 1 bar and 28°C. If the relative humidity is found to be 30%, determine: [8]
- Dew point temperature of the air.
 - The Specific volume of each constituent.
 - The Specific Humidity.
- c) State and explain the factors which govern the optimum effective temperature? [4]

OR

- Q6)** a) Define the following terms: [4]
- Grand sensible heat factor (GSHF)
 - Infiltration and Ventilation air
 - Room sensible heat factor (RSHF)
 - Outside air total heat load (OATHL)
- b) Mechanical Engineering Departmental office of 25 person's seating capacity is provided with summer air-conditioning system with the following data: [8]
- | | |
|----------------------------------|-------------------------------|
| Outside Condition | 34°C DBT and 28°C WBT |
| Inside conditions: | 24°C DBT and 50% RH |
| Volume of air supplied | 0.4m ³ /min/person |
| Sensible heat load in the office | 125600 kJ/hr |
| Latent heat load in the office | 42000 kJ/hr |
- Find the sensible heat factor of the plant?
- c) Write a note on "Indoor air quality"? [4]

- Q7)** a) Explain with neat Sketch any two of the following Air-conditioning system: [8]
- Year round air-conditioning system.
 - Industrial air-conditioning system.
 - Comfort air-conditioning system.
- b) Explain the working of Thermostatic Expansion valve with the help of neat sketch? Enumerate its advantages and disadvantages over other type of expansion devices. [8]

OR

Q8) a) What do you mean by Evaporators? Explain the functioning of flooded evaporators with the help of neat sketch? [8]

b) Write a note on “operating and safety controls” of Refrigeration and Air-Conditioning Systems. List out minimum 08 safety controls of the system? [8]

Q9) a) Why “Ducts” are arranged in an air-conditioning system? Explain either “Radial Perimeter duct system” or “Extended plenum duct system” with neat sketch? [6]

b) A 12 cm long duct passes air at the rate of 1.2 m³/s. If the friction factor is 0.0048, calculate the pressure drop in the following cases: [6]

i) When the duct is circular of diameter 28 cm;

ii) When the duct is 28 cm square section.

c) Explain the following types of supply air outlets? [6]

i) Grille outlets

ii) Ceiling diffuser outlets

iii) Slot diffuser outlets.

OR

Q10) a) What do you mean by equivalent diameter of rectangular duct? Derive an expression for equivalent diameter of rectangular duct for the same air flow as

$$D = 1.265 \left[\frac{a^3 b^3}{a + b} \right]^{0.2}$$
 where, ‘a’ and ‘b’ are the sides of the rectangular duct. [6]

b) Explain any two of the following air distribution systems: [8]

i) Ejector system

ii) Downward system

iii) Upward system

c) State the principles of air distribution? [4]

